

Firewall Configurations Theory (1 hour):

Basics of firewalls and ufw/iptables in Linux. Creating rules for inbound and outbound traffic.

Practical (1 hour): Configure ufw to allow SSH and block other services. Task: Document the configured firewall rules.

TASKS

1. Install ufw on your Linux system (if not already installed) and enable it, then verify its status using the appropriate commands.

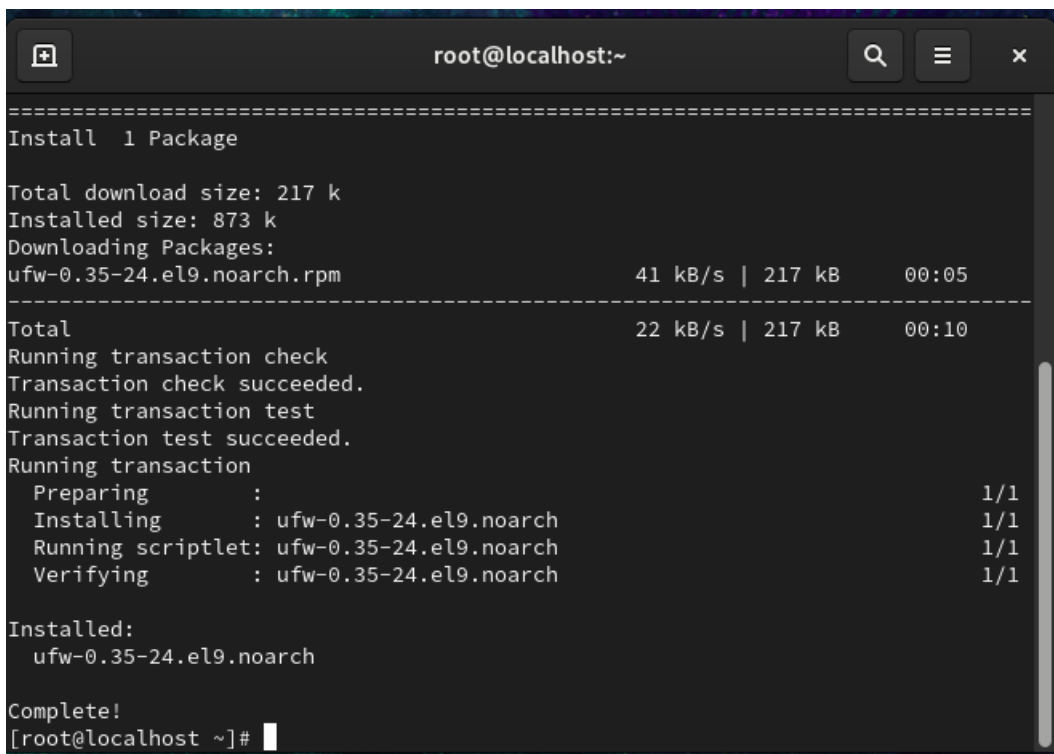
Install UFW and Verify

Step 1: Install EPEL Repository

```
dnf install epel-release -y
```

Step 2: Install UFW

```
dnf install ufw -y
```



```
root@localhost:~
=====
Install 1 Package
=====
Total download size: 217 k
Installed size: 873 k
Downloading Packages:
ufw-0.35-24.el9.noarch.rpm                                41 kB/s | 217 kB    00:05
-----
Total                                                    22 kB/s | 217 kB    00:10
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : ufw-0.35-24.el9.noarch         1/1
  Running scriptlet: ufw-0.35-24.el9.noarch        1/1
  Verifying      : ufw-0.35-24.el9.noarch         1/1

Installed:
  ufw-0.35-24.el9.noarch

Complete!
[root@localhost ~]#
```

Step
3:

Enable UFW Service

```
systemctl enable ufw
systemctl start ufw
```

Step 4: Check Status

```
ufw status
```

If inactive, enable it:

```
ufw enable
```

Type:

y

Check again:

```
ufw status verbose
```

```
bash: y: command not found...
[root@localhost ~]# ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22/tcp (SSH) ALLOW IN Anywhere
224.0.0.251 5353/udp (mDNS) ALLOW IN Anywhere
22/tcp (SSH (v6)) ALLOW IN Anywhere (v6)
ff02::fb 5353/udp (mDNS) ALLOW IN Anywhere (v6)
[root@localhost ~]#
```

2. Configure ufw to allow incoming SSH connections only on port 22 and block all other incoming connections.

Allow SSH Only

Step 1: Reset Rules (Optional)

```
ufw --force reset
```

Step 2: Set Default Policies

```
ufw default deny incoming
ufw default allow outgoing
```

Step 3: Allow SSH (Port 22)

ufw allow 22/tcp

or

ufw allow ssh

Step 4: Verify

ufw status numbered

```
[root@localhost ~]# ufw status numbered
Status: active

      To Action From
      --
[ 1] SSH ALLOW IN Anywhere
[ 2] 224.0.0.251 mDNS ALLOW IN Anywhere
[ 3] 22/tcp ALLOW IN Anywhere
[ 4] 22 ALLOW IN Anywhere
[ 5] SSH (v6) ALLOW IN Anywhere (v6)
[ 6] ff02::fb mDNS ALLOW IN Anywhere (v6)
[ 7] 22/tcp (v6) ALLOW IN Anywhere (v6)
[ 8] 22 (v6) ALLOW IN Anywhere (v6)

[root@localhost ~]#
```

3. Add a rule in ufw to allow outgoing HTTP and HTTPS traffic (ports 80 and 443), but block all other outbound connections.
Hint: Use 'ufw allow out' and 'ufw deny out' commands to control outbound traffic.

Allow Outgoing HTTP and HTTPS

Step 1: Allow HTTP

ufw allow out 80/tcp

Step 2: Allow HTTPS

ufw allow out 443/tcp

Step 3: Block All Other Outgoing

ufw default deny outgoing

❑ **Important:** This may block your internet connection. If this is only for the assignment, take the screenshot and then restore it later.

Step 4: Allow HTTP/HTTPS Again

ufw allow out 80/tcp

ufw allow out 443/tcp

Step 5: Verify

ufw status verbose

```
snipping during existing rule (v6)
[root@localhost ~]# ufw status verbose
Status: active
Logging: on (low)
Default: allow (incoming), deny (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22/tcp (SSH) ALLOW IN Anywhere
224.0.0.251 5353/udp (mDNS) ALLOW IN Anywhere
22/tcp ALLOW IN Anywhere
22 ALLOW IN Anywhere
22/tcp (SSH (v6)) ALLOW IN Anywhere (v6)
ff02::fb 5353/udp (mDNS) ALLOW IN Anywhere (v6)
22/tcp (v6) ALLOW IN Anywhere (v6)
22 (v6) ALLOW IN Anywhere (v6)

80/tcp ALLOW OUT Anywhere
443/tcp ALLOW OUT Anywhere
80/tcp (v6) ALLOW OUT Anywhere (v6)
443/tcp (v6) ALLOW OUT Anywhere (v6)

[root@localhost ~]#
```

4. List all currently active ufw rules on your system and save the output to a file named `firewall_rules.txt`.

Save Firewall Rules

Save the rules:

ufw status numbered > firewall_rules.txt

Verify:

cat firewall_rules.txt

```

[root@localhost ~]# ufw status numbered > firewall_rules.txt
[root@localhost ~]# cat firewall_rules.txt
Status: active

      To                Action      From
      --                -
[ 1] SSH                ALLOW IN   Anywhere
[ 2] 224.0.0.251 mDNS    ALLOW IN   Anywhere
[ 3] 22/tcp              ALLOW IN   Anywhere
[ 4] 22                  ALLOW IN   Anywhere
[ 5] 80/tcp              ALLOW OUT   Anywhere          (out)
[ 6] 443/tcp             ALLOW OUT   Anywhere          (out)
[ 7] SSH (v6)            ALLOW IN   Anywhere (v6)
[ 8] ff02::fb mDNS       ALLOW IN   Anywhere (v6)
[ 9] 22/tcp (v6)         ALLOW IN   Anywhere (v6)
[10] 22 (v6)             ALLOW IN   Anywhere (v6)
[11] 80/tcp (v6)         ALLOW OUT   Anywhere (v6)      (out)
[12] 443/tcp (v6)        ALLOW OUT   Anywhere (v6)      (out)

[root@localhost ~]#

```

5. Write a short explanation (3-4 lines) describing how your current ufw configuration would impact running a local web server (like a development server for a Zomato-style food delivery app).

The firewall allows only SSH incoming connections on port 22, improving system security. Outgoing web traffic is limited to HTTP (80) and HTTPS (443), which is sufficient for accessing websites and software updates. A local web server can still serve content if the required ports are allowed, while unnecessary network access remains blocked.

```

[root@localhost ~]# ufw status numbered > firewall_rules.txt
[root@localhost ~]# cat firewall_rules.txt
Status: active

      To                Action      From
      --                -
[ 1] SSH                ALLOW IN   Anywhere
[ 2] 224.0.0.251 mDNS    ALLOW IN   Anywhere
[ 3] 22/tcp              ALLOW IN   Anywhere
[ 4] 22                  ALLOW IN   Anywhere
[ 5] 80/tcp              ALLOW OUT   Anywhere          (out)
[ 6] 443/tcp             ALLOW OUT   Anywhere          (out)
[ 7] SSH (v6)            ALLOW IN   Anywhere (v6)
[ 8] ff02::fb mDNS       ALLOW IN   Anywhere (v6)
[ 9] 22/tcp (v6)         ALLOW IN   Anywhere (v6)
[10] 22 (v6)             ALLOW IN   Anywhere (v6)
[11] 80/tcp (v6)         ALLOW OUT   Anywhere (v6)      (out)
[12] 443/tcp (v6)        ALLOW OUT   Anywhere (v6)      (out)

[root@localhost ~]#

```